Mechanical Properties of Ceramics and Glasses

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Aims of session & Learning Outcomes

Aims

- Discuss mechanical properties of ceramics
- Explain fracture mechanism for ceramics
- Discuss the strain-stress behavior of ceramics

Learning Outcomes

- Identify characteristic of ceramics that affect their mechanical behavior

Mechanical behaviour

There are 3 important mechanical properties

- Elastic properties – modulus of elasticity
- Plastic properties – yield or ultimate strength
- Fracture properties – toughness

Which are important for ceramic materials?

Properties of ceramics

	E GPa	$\sigma_{compressive}$ MPa	*Kc* MPa√m	Tm K	Thermal conductivity W/m/K
Soda-lime glass	74	1000	0.7	~1000	1
Porcelain	70	350	1	~1000	1
Al_2O_3	380	3000	3-5	2323	25.6
SiC	410	2000	0.5	3110	84
Si_3N_4	310	1200	4	2173	17
ZrO_2	200	2000	4-12	2843	1.5
Diamond	1050	5000	-	-	70

Ashby and Jones II Chapter 17

Elastic properties

Stress – strain behavior

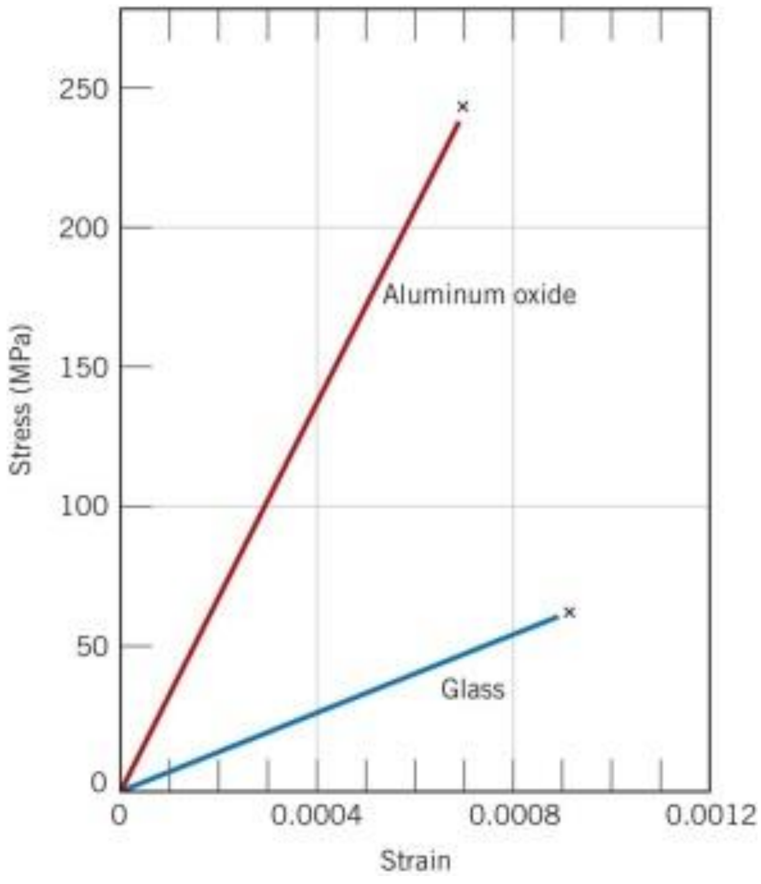


Fig_14_10 Callister, 9e

- Elastic modulus $\sigma = E\varepsilon$
- The relationship between stress and strain is linear (linear elastic solid)
- Slope = modulus of elasticity

  Glass & porcelain ~ 70 – 85 GPa

  Engineering ceramics ~ 200-650 GPa
- Strong atomic bonds
- No plastic deformation is observed

- **Brittle elastic failure**

Fracture mechanics

Toughness

measured by the **Stress Intensity Factor**

$$K_c = Y\sigma\sqrt{\pi a}$$

where σ = the applied stress
a = the crack length
Y = shape factor depending on the geometry of the crack

K_c = the critical value for crack propagation
= the material's resistance to crack propagation

From this equation we can see that:

$$\sigma_{fs} = \frac{K_c}{Y\sqrt{\pi a}}$$

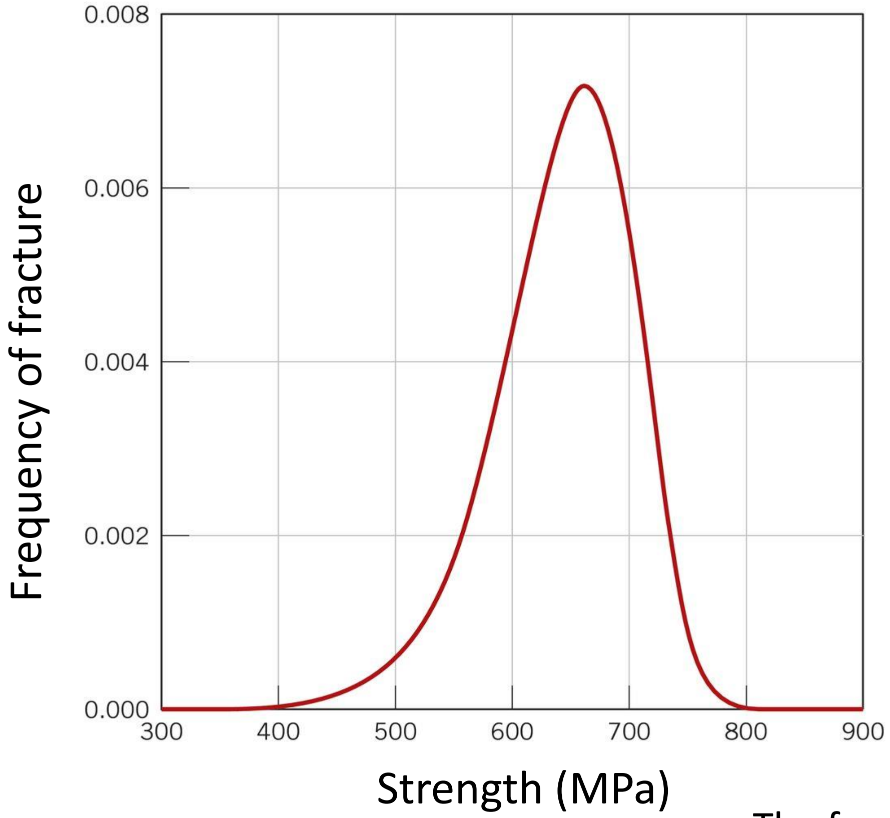
The **fracture stress**, σ_{fs}, depends on the 1/ √crack length
Large flaws = lower strength

For ceramics and glasses = strength depends on existing flaws and defects

Distribution of strength

Repeated measurements give a distribution of strength values:

- Depends on flaw population in each test specimen
- Larger flaws = lower strength
- Edges of specimen critical
- For random distribution of flaws the distribution of strengths is skewed because $\sigma_{fs} \propto 1/\sqrt{a}$



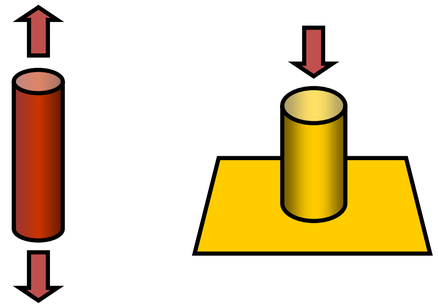
The frequency distribution of observed fracture strengths for a silicon nitride material.

Strength values depend on specimen size and surface finish

Fig. 14.5, *Callister & Rethwsich 9e.*

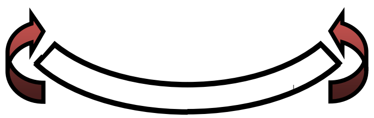
Mechanical testing of ceramics

Tensile testing problematic:

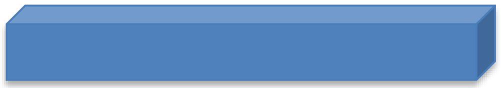


- Difficult to grip
- Stiff materials – difficult to load axially
- Very small elastic strain – difficult to measure
- Compression testing easier

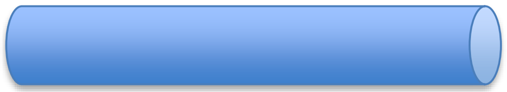
Flexural testing:



- No gripping problem
- Displacement measurement maximised
- Stress at tensile surface



- **Rectangular bar** has edges = edge flaws

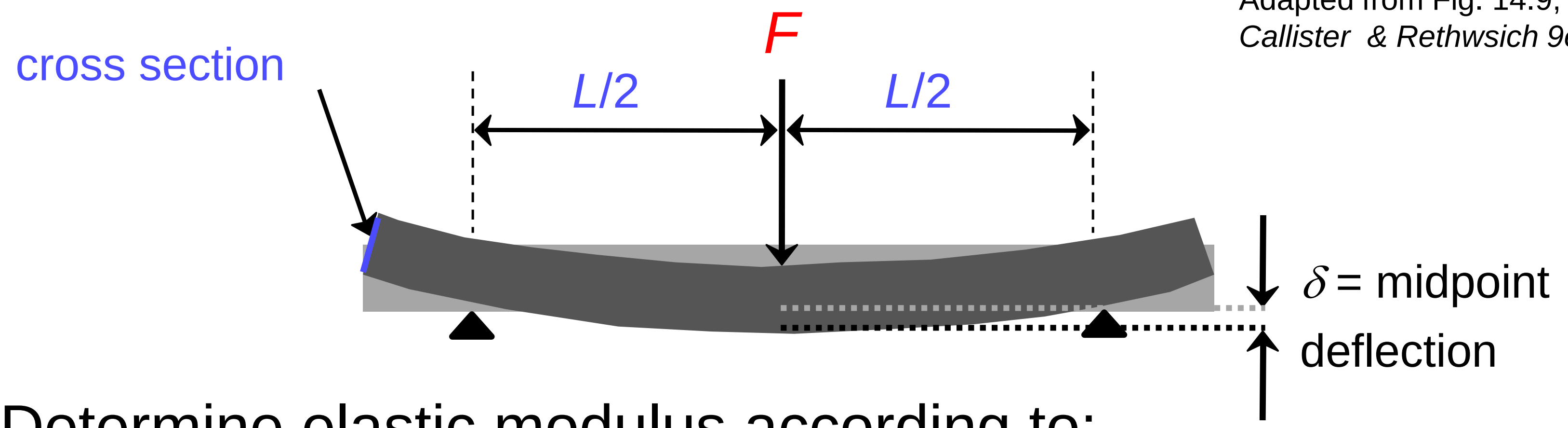


- **Circular bar** has no edges = surface flaws
  (smaller stressed volume)

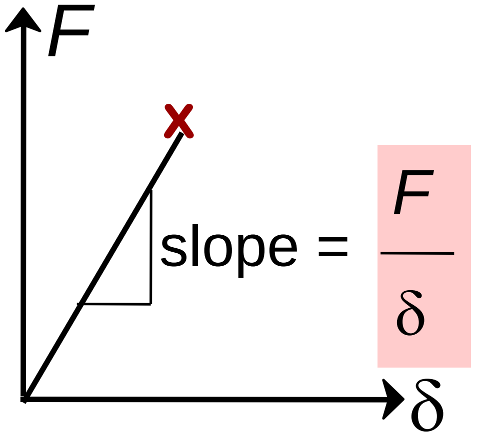
Measuring Elastic Modulus

3-Point Bend Testing often used.

Adapted from Fig. 14.9, *Callister & Rethwsich 9e.*

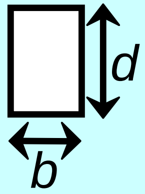


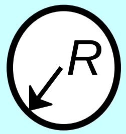
- Determine elastic modulus according to:



linear-elastic behavior

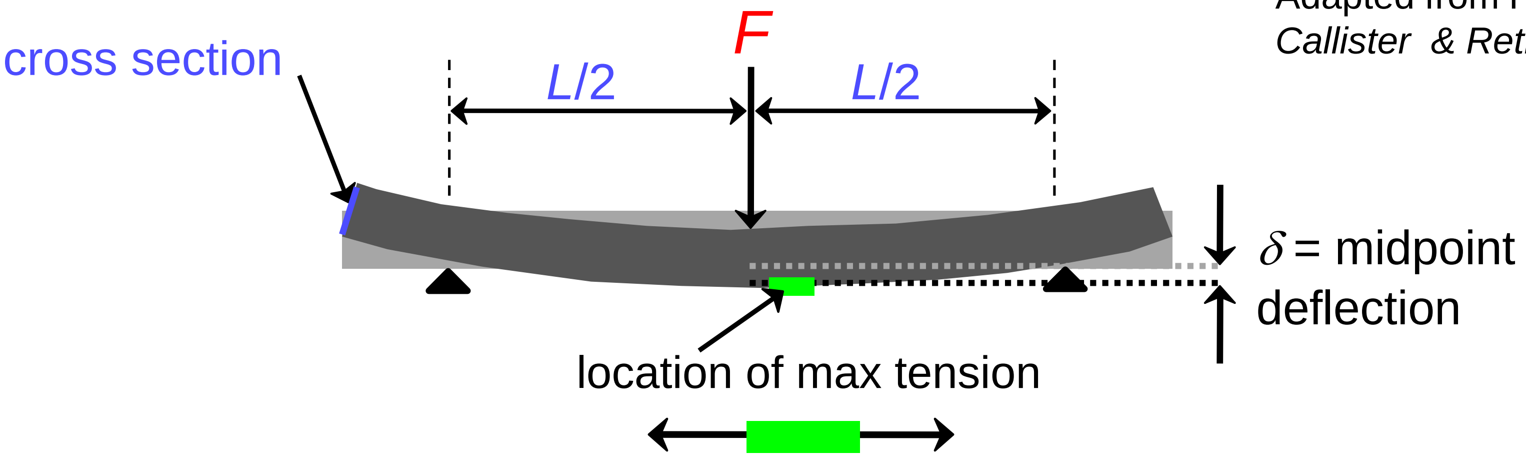
$$E = \frac{F}{\delta}\,\frac{L^3}{4bd^3} = \frac{F}{\delta}\,\frac{L^3}{12\pi R^4}$$





Measuring Strength

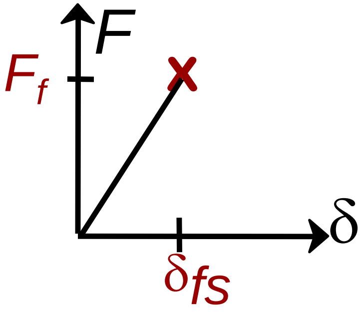
3-point bend test to measure room *T* strength.

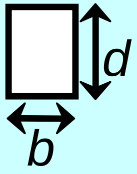


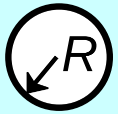
Adapted from Fig. 14.9, *Callister & Rethwsich 9e.*

- Flexural strength:

$$\sigma_{fs} = \frac{1.5 F_f L}{b d^2} = \frac{F_f L}{\pi R^3}$$







Typical values:

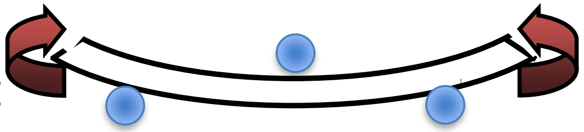
Material	σ_{fs} (MPa)	E (GPa)
Si nitride	250-1000	304
Si carbide	100-820	345
Al oxide	275-700	393
glass (soda)	69	69

Data from Table 12.5, *Callister 7e.*

Better test methods

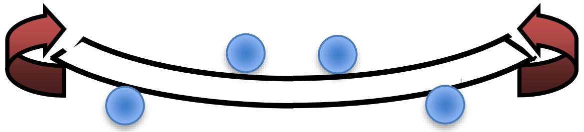
3 point bending:

- Easy to set up
- Displacement measured from central contact
- Can create stress concentration in centre



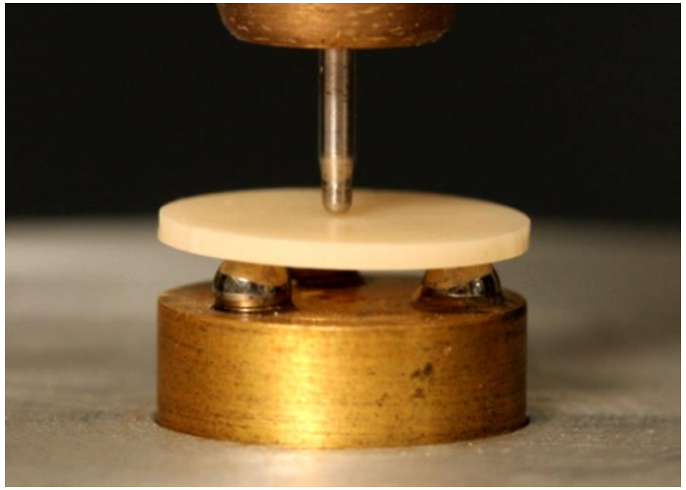
4 point bending:

- Circular bending in central section
- Tensile stress constant over large area
- Displacement measurement more difficult



Bi-axial flexure:

- 2D flexure of disc between supports and central load point
- No edges
- Large surface area in uniform tension



Plasticity or cracking?

Strength dominated by flaws

- Failure by crack propagation in tension
- Failure in compression also by cracking
- From flaws in the material from processing
- and specimen preparation

Plasticity in ceramics and glass:

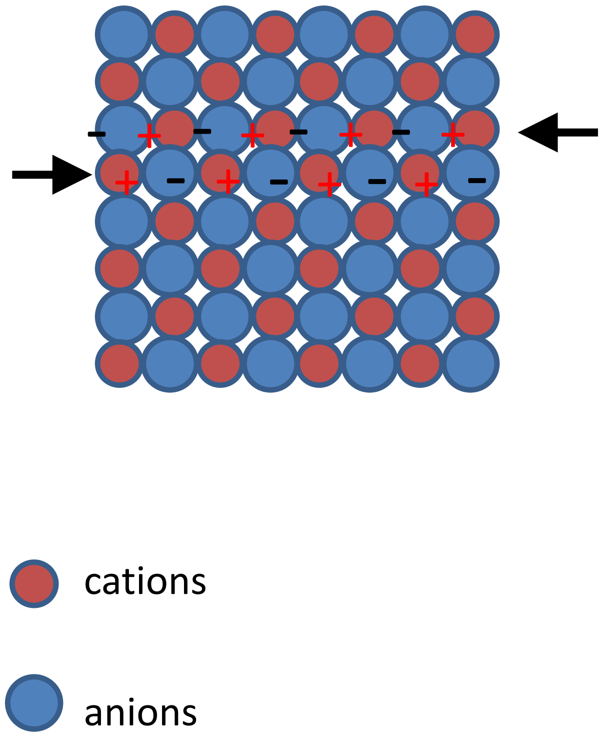
- Plasticity only in confined geometry e.g. hardness test

Plasticity in **crystalline ceramics**
by dislocation motion (limited slip systems)
and by twinning
Charge balance must be maintained in slip direction

Plasticity in glasses by **shear faults**

Ceramics – plastic deformation

- Deformation in limited number of slip systems



- Ceramic crystals can deform by dislocation slip as in metal crystals

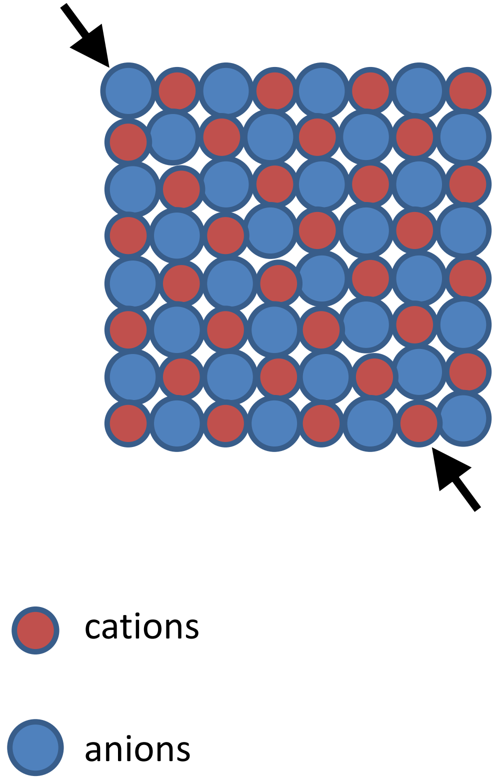
- Ionic solids must maintain ‘charge balance’

- So that some slip directions are forbidden as it would bring same sign charges together and they repel each other

  NOT PERMITTED

Ceramics – plastic deformation

- Deformation in limited number of slip systems



- Ceramic crystals can deform by dislocation slip as in metal crystals
- Ionic solids must maintain 'charge balance'
- So that some slip directions are forbidden as it would bring same sign charges together and they repel each other
  NOT PERMITTED
- Other slip directions are permitted and maintain the charge balance in the crystal

The limited slip systems available make ceramic materials more difficult to plastically deform = more brittle

Fractography of ceramics

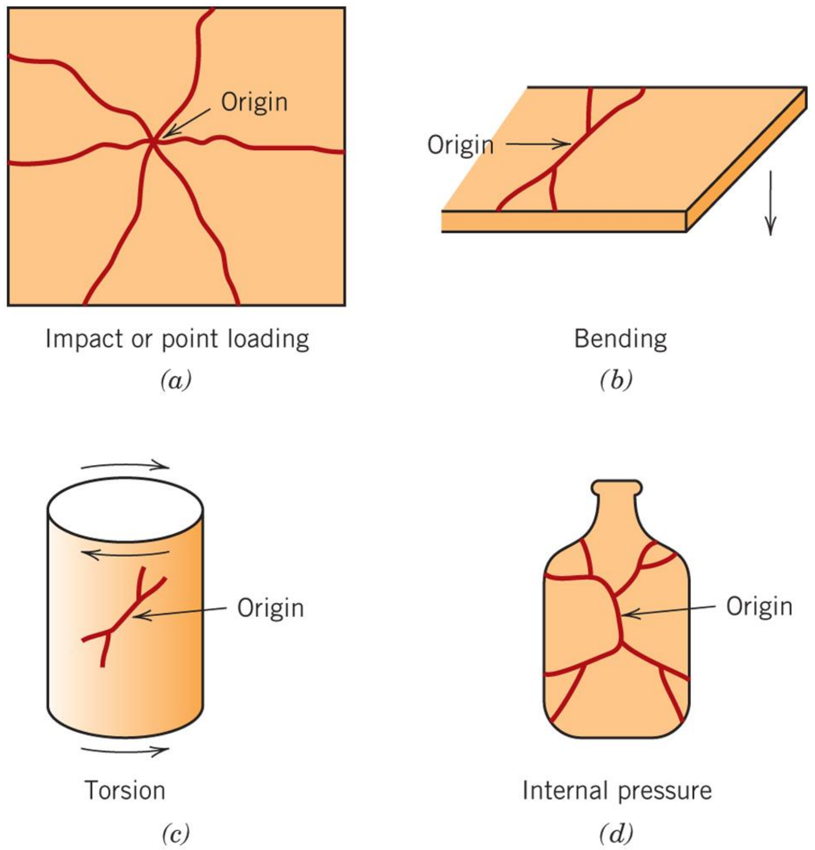


Fig. 14.6, *Callister & Rethwsich 9e.*

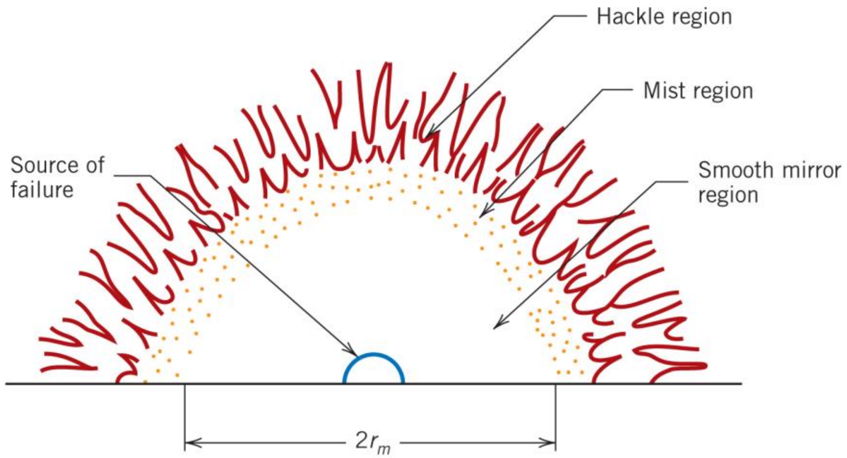
- Analysis of failure of ceramics
- Path of fracture propagation and microstructure of fracture
- The point of nucleation or crack initiation can be traced

The **origin of the fracture** (loading point) and **type of loading**

(impact, bending, torsion, pressure)

can be determined from the crack pattern – **used in police investigations**

Fractography of ceramics



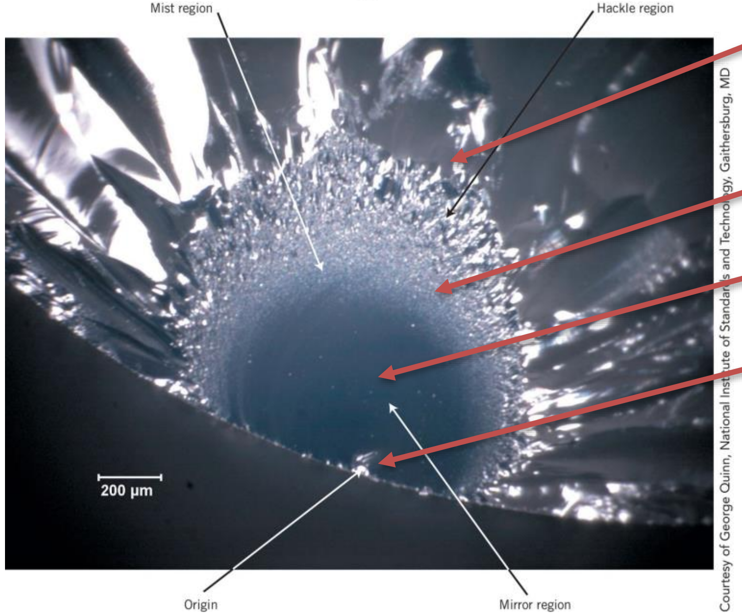


Fig. 14.7-8, *Callister & Rethwsich 9e.*

- The point of nucleation or crack initiation can be traced
- Cracks typically will exhibit the following regions:
    - Hackle: striations or lines that radiate away from the source
    - Mist: faint, annular region
    - Mirror: smooth
    - Origin: at centre of circular crack

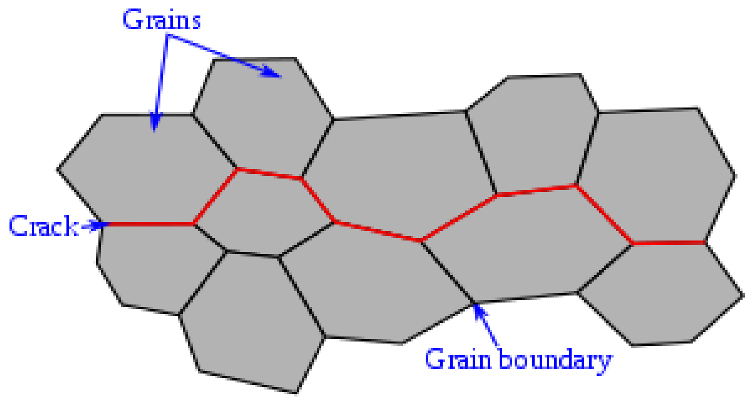
Microscopy examination of fracture surface can determine the fracture origin and initial flaw size

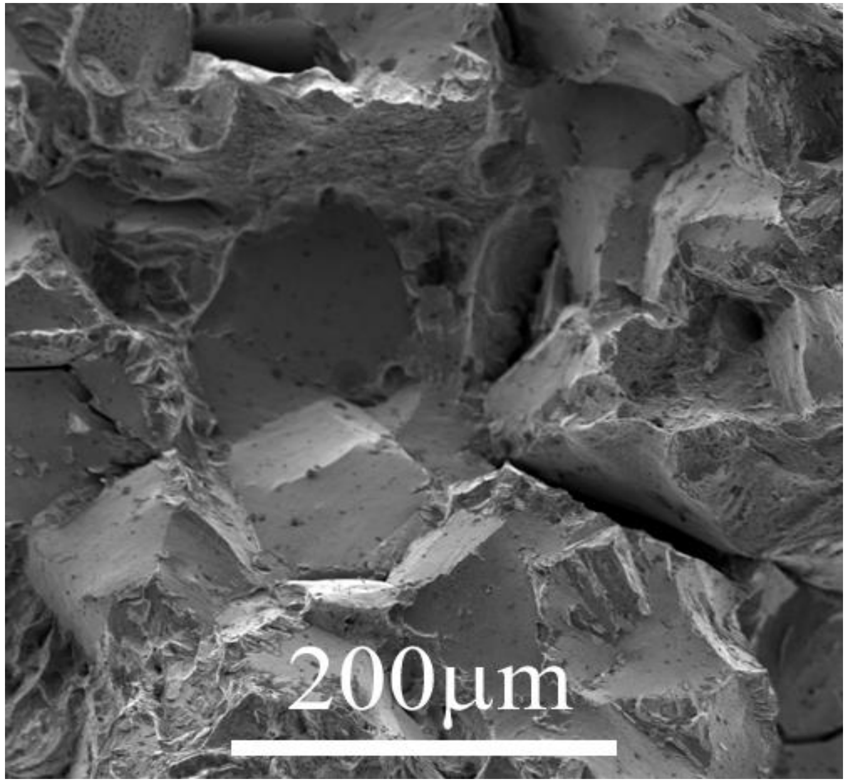
Fractography of ceramics

Fracture in brittle crystalline solids is often intergranular (between the crystal grains)

The crystal grain boundaries are **relatively weak** because of:

- Insoluble **second phases** (from impurities) are left at grain boundaries
- **Porosity** at grain boundaries from sintering

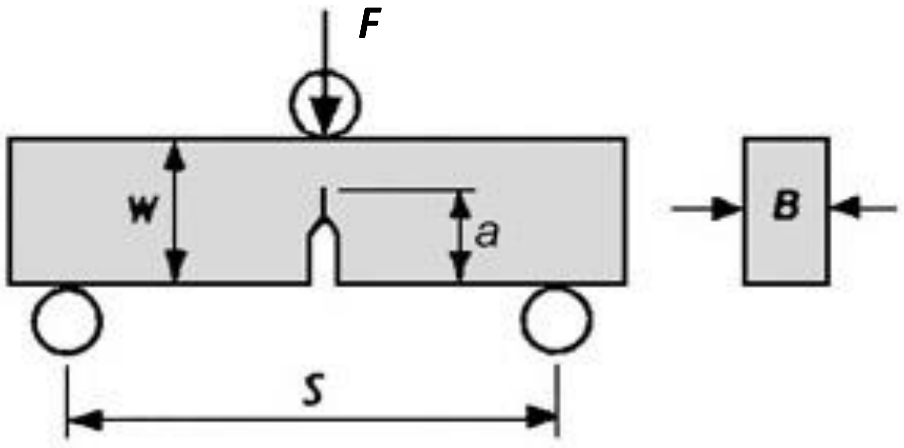




DoITPoMs micrograph library

Fracture testing

To determine fracture toughness



Single edge notch beam specimen

Special test shape:
Requires test specimen made with a sharp crack in it

Increase the force, *F*, until the sample suddenly breaks, F_c

Calculate K_{IC} from formula below:

$$K_{IC} = \frac{F_c\, S}{BW^{3/2}} f\left(\frac{a}{W}\right)$$

$$f\left(\frac{a}{W}\right) = \frac{3\left(\frac{a}{W}\right)^{\frac{1}{2}}\left[1.99 - \frac{a}{W}\left(1-\frac{a}{W}\right)\left\{2.15 - 3.93\left(\frac{a}{W}\right) + 2.7\left(\frac{a}{W}\right)^2\right\}\right]}{2\left(1+2\frac{a}{W}\right)\left(1-\frac{a}{W}\right)^{\frac{3}{2}}}$$

Values for K_{IC} typically 0.5 – 10 MPa√m for ceramics and glass

Porosity

Porosity in ceramics affects both the elastic modulus and strength:

Elastic modulus $E = E_0 (1 - 1.9P + 0.9P^2)$

Flexural strength $\sigma_{fs} = \sigma_0 \exp(-nP)$

where P = volume fraction porosity
E_0 is elastic modulus for $P = 0$
σ_0 and n are constants

Strength reduces strongly with porosity since the pores can act as stress concentrations to initiate fracture

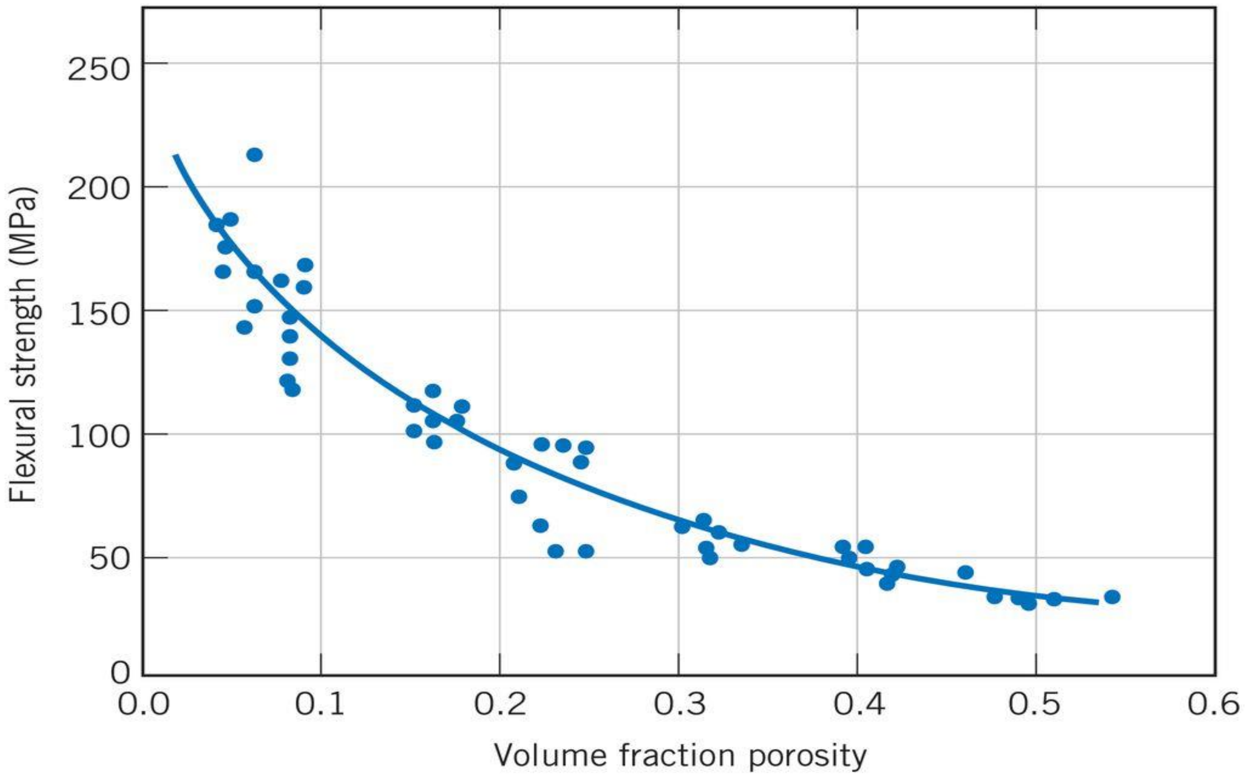


Fig. 14.13, *Callister & Rethwsich 9e.*

Summary

Mechanical properties of ceramics and glasses

- Linear elastic – limited plasticity
- Fracture strength depends of flaws in the material
- Distribution of fracture strength because of flaw distribution
- Test and specimen preparation dependent
- Strength and toughness often confused for ceramics since both depend on cracks in the material
- Porosity and impurities from processing also affect mechanical properties